

PHASE 8 — LIMITATIONS

Consolidated from the phase report, per Master Prompt Section 16.

1. No flood, environmental, or hazard data exists anywhere in this platform

Checked across Phase 1's full table set — no elevation data, no floodplain boundaries, no environmental monitoring readings, no climate-risk indicators of any kind. `geo_risk.compute_geo_risk()` therefore returns `flood_risk = {"status": "INSUFFICIENT DATA"}` and `environmental_risk = {"status": "INSUFFICIENT DATA"}` for **all 90 of 90 localities**, unconditionally — verified by a dedicated regression test across a spread of locality IDs. This is a genuine data gap in the source platform, not a limitation of this phase's logic; if a real hazard dataset becomes available, `data_loader.py` and `geo_risk.py` are the two places to wire it in.

2. Infrastructure coverage is sparse

209 infrastructure records across 90 localities (2.32 per locality on average). 65.6% of localities are missing at least one of the three tracked categories (transit, social, commercial), so their accessibility and infrastructure sub-scores are PASS WITH LIMITATIONS rather than PASS — the score is computed from whatever categories *are* present, with weights renormalized, rather than penalizing the missing categories as if they scored zero.

3. Location Score weights are a documented default, not a fitted model

`location_score.WEIGHTS` (connectivity 15%, safety 15%, infrastructure 15%, development 15%, commercial 10%, accessibility 15%, market strength 15%) reflect a business-judgement starting point. No historical investment-outcome dataset exists in this platform (e.g., “which localities actually performed best for investors”) to fit these weights against. A real deployment should have the business explicitly review and, if desired, replace these before relying on the composite for investment decisions.

4. The distance-decay formula for accessibility is a simple, documented choice

`accessibility._proximity_weight(distance_km) = 1/(1+distance_km)` is monotonic and bounded but not fitted to any real travel-time, transit-frequency, or traffic data — none exists in this dataset. A more realistic model (e.g. actual transit travel times) would require external data this platform does not have.

5. Infrastructure delivery-risk and concentration-risk are proxy indicators

Both are computed directly from real `core.infrastructure` fields (`expected_completion`, `impact_score`), but they describe *this dataset's* infrastructure composition — they are not calibrated probabilities that any specific project will slip or that any specific locality will suffer a specific loss. This mirrors the honesty standard Phase 7 set for its Monte Carlo UNCALIBRATED status: a real number, honestly computed, but not a validated real-world probability.

6. expected_completion dates are taken at face value

No independent verification exists (in this platform) that a project marked “completed” (past `expected_completion`) was actually delivered on schedule, on budget, or as originally specified. The delivery-risk proxy can only measure *scheduling* status as recorded, not construction-quality or delivery-fidelity risk.

7. Market-strength normalization was fixed once during development — worth re-checking if source data changes

The original cap-based formula ($\text{ratio}=1.0 \rightarrow 100$) saturated 78% of localities at the ceiling because this dataset's demand/supply ratios and absorption rates run structurally above 1.0. The fix uses P5/P95 anchors computed from the actual 90-locality distribution at build time. **If the underlying market_monthly data is regenerated or substantially changed**, these anchors (`RATIO_P5=1.06`, `RATIO_P95=4.89`, `ABS_P5=0.48`, `ABS_P95=6.21`, hardcoded in `location_score.py`) should be recomputed — they are data-driven constants, not universal ones, and the regression test (`test_market_strength_not_saturated_across_portfolio`) will catch re-saturation if they drift out of sync with new data.

8. No live PostgreSQL was available in this sandbox

Same situation as Phases 3, 5, 6, and 7. All orchestrators default to `--source db` (the production path) but were run here with `--source csv`, reading the identical rows Phase 1 loaded. Recommend re-running with `--source db` before Phase 9 integration.

9. City-level differentiation is weak in this composite

Mean Location Score by city ranges narrowly from 52.6 to 54.9 across all 6 cities — this dataset's city-level generation doesn't produce strong city-to-city separation in the composite. Locality-to-locality variation (37.6 to 70.5) is much more informative than city-to-city variation; a caller should not expect this score to answer “which city is best” with much confidence.